THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 6: **Heat Transfer in Nature** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Water in a beaker heated from below gets heated by: 1

(a) convection currents **(b)** evaporation
(c) radiation only **(d)** conduction only

2. Woollen clothes keep us warm because wool: 1

(a) is a good conductor **(b)** produces heat
(c) is shiny **(d)** traps air, a poor conductor

3. Transfer of heat without any medium is: 1

(a) conduction **(b)** convection **(c)** evaporation **(d)** radiation

4. Which of the following is heated mostly by conduction? 1

(a) the Earth by Sun **(b)** a metal rod in a flame
(c) water in a pot **(d)** room air above a heater

5. A thermos flask keeps tea hot by: 1

(a) reducing conduction, convection and radiation
(b) producing heat
(c) increasing conduction
(d) using an iron body

6. Heat from the Sun reaches the Earth by: 1

(a) convection **(b)** conduction **(c)** radiation **(d)** sublimation

7. Which of the following is heated mostly by radiation? 1

(a) a person sitting in sunlight **(b)** water in a kettle
(c) air near a heater **(d)** a spoon in tea

8. The silvered walls of a thermos flask reduce: 1

(a) convection only **(b)** all heating **(c)** radiation **(d)** conduction only

9. Which of these is a good conductor of heat? 1

(a) plastic **(b)** wood **(c)** rubber **(d)** aluminium

10. The sea breeze blows during the: 1

(a) night **(b)** evening only **(c)** midnight **(d)** day

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do we wear woollen clothes in winter? 2

12. Why are the handles of utensils made of plastic or wood? 2

13. What is convection? Give an example. 2

14. What is a sea breeze? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how heat from the Sun reaches the Earth. 3

16. Describe the construction of a thermos flask. 3

17. Explain why glass windows are used in greenhouses. 3

OR

Explain sea breeze with the help of convection. 3

18. Give three examples each of conductors and insulators. 3

OR

Why does a room get warmer when an oil heater is switched on, even at a distance? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe sea and land breeze and draw their directions in words. 5

OR

Describe the use of insulators in daily life with five examples. 5

20. Explain how clothes of different colours and materials help us in different seasons. 5

OR

Explain why hot air balloons rise. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND